ITP Player Database — Performance Benchmark Sources

Evidence-based benchmarks from European football academy research Date: March 2026

Overview

This document provides the scientific basis for all performance test benchmarks used in the ITP Player Database. Each test section includes published reference data, derived threshold values per age group, confidence ratings, and full citations. The goal is to ensure every threshold shown to coaches, scouts, and players is grounded in peer-reviewed research from European football academies.

Methodology

Thresholds were derived using the following approach:

1. **Primary sources** — Large-sample studies from German Bundesliga *Nachwuchsleistungszentren* (NLZ) were prioritized, as they are closest in context to 1. FC Koln's ITP program.
2. **Supporting sources** — Data from other top European academies (Spanish La Liga, Portuguese Primeira Liga, Austrian Bundesliga, Belgian Pro League, Aspire Academy) were used to validate and cross-reference.
3. **Threshold bands** — "Poor" represents below-average academy level, "Average" represents the mean for that age group, "Good" represents above-average / starter-level, and "Elite" represents top-tier / professional-level performance.
4. **Age groups** — U-17 (15-16 years), U-19 (17-18 years), U-21 (19-20 years).

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1. Countermovement Jump (CMJ)

Published Data

Study	N	Age Group	Level	Mean +/- SD (cm)
Hoppe et al. (2020)	1,134	U17	German Bundesliga NLZ	40.3 +/- 1.3
Hoppe et al. (2020)	1,134	U17 starters	German Bundesliga NLZ	40.5
Hoppe et al. (2020)	1,134	U17 non-starters	German Bundesliga NLZ	40.0
Hoppe et al. (2020)	1,134	U19	German Bundesliga NLZ	41.6 +/- 1.2
Hoppe et al. (2020)	1,134	U19 starters	German Bundesliga NLZ	42.1
Hoppe et al. (2020)	1,134	U19 non-starters	German Bundesliga NLZ	41.0
Hoppe et al. (2020)	1,134	U21	German Bundesliga NLZ	45.2 +/- 2.5
Hoppe et al. (2020)	1,134	U21 starters	German Bundesliga NLZ	47.4
Hoppe et al. (2020)	1,134	U21 non-starters	German Bundesliga NLZ	44.2
Nikolaidis (2014)	—	U16	Greek youth	36.1 +/- 6.4
Nikolaidis (2014)	—	U18	Greek youth	40.8 +/- 5.5

Rebelo et al. (2013)	—	U19	Portuguese 1st Division	38.9 +/- 4.6
Los Arcos et al. (2015)	—	U19	Spanish La Liga	~45.4 +/- 4.3
Lopez-Segovia et al. (2020)	—	U21	Spanish La Liga reserve	45.2 +/- 4.2

Threshold Derivation

The Hoppe et al. (2020) dataset from German Bundesliga NLZ serves as the primary reference — it is the largest and most contextually relevant study for 1. FC Koln's ITP program. Supporting European data validates the range.

- **"Poor"** is set approximately 1 SD below the NLZ mean, roughly matching lower-tier academy players or non-elite youth.
- **"Average"** represents the approximate NLZ mean for that age group.
- **"Good"** is set above the NLZ mean, roughly matching starter-level performance.
- **"Elite"** represents the upper end of the European academy distribution, matching professional-transition performance.

Thresholds (higher is better)

Age Group	Poor	Average	Good	Elite
U-17	<34 cm	34-38 cm	39-42 cm	43+ cm
U-19	<36 cm	36-40 cm	41-44 cm	45+ cm
U-21	<38 cm	38-42 cm	43-47 cm	48+ cm

Confidence: HIGH — Large German NLZ sample (Hoppe et al., n=1,134), corroborated by multiple European studies. This is the most well-researched test in football academy literature.

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2. Broad Jump (Standing Long Jump)

Published Data

Study	N	Age Group	Level	Mean +/- SD (cm)
Vaeyens et al. (2006)	—	U16	Belgian elite	201.5 +/- 13.6
Nowak et al. (2025)	—	Age 15 (P50)	Polish academy	232
Nowak et al. (2025)	—	Age 16 (P50)	Polish academy	235
Deprez et al. (2015)	—	U14-U18	Belgian longitudinal model	Developmental curve

Threshold Derivation

Broad Jump data in European football academies is considerably sparser than CMJ data. The Belgian (Vaeyens, Deprez) and Polish (Nowak) studies provide the best available reference points. For U-19 and U-21, thresholds were estimated from developmental trajectory models and cross-referenced with NCAA Division I data (~241 cm as a college-level reference).

The wide variation across studies (Belgian U16 mean of 201.5 vs. Polish P50 of 235 at age 16) likely reflects differences in testing protocol, sample selection, and population. Thresholds were set conservatively to account for

this.

Thresholds (higher is better)

Age Group	Poor	Average	Good	Elite
U-17	<190 cm	190-210 cm	211-230 cm	231+ cm
U-19	<200 cm	200-220 cm	221-240 cm	241+ cm
U-21	<210 cm	210-230 cm	231-250 cm	251+ cm

Confidence: MEDIUM (U-17), LOW (U-19 / U-21) — Limited European football-specific data. U-19 and U-21 thresholds are partially estimated from developmental curves and cross-sport references. More Bundesliga NLZ data would strengthen these values.

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3. 5m Sprint

Published Data

Study	N	Age Group	Level	Mean +/- SD (s)
Gonaus et al. (2019)	261-998 per age	U15-U19	Austrian Bundesliga NLZ	Age-specific norms
Bujnovsky et al. (2019)	—	U17	Czech First League	~1.10-1.16
Ruf et al. (2024)	—	U11-U19	TSG Hoffenheim (Bundesliga NLZ)	5m centile data

Threshold Derivation

5m sprint data is relatively scarce in football literature because many academies test 10m or 20m rather than 5m as their short-distance benchmark. The Gonaus et al. (2019) Austrian NLZ dataset is the most powerful source (hundreds of athletes per age group). The Ruf et al. (2024) Hoffenheim data is particularly valuable as a direct Bundesliga NLZ reference.

Performance plateaus around age 16-17 for the 5m distance, meaning U-19 and U-21 thresholds are relatively close together. Best estimates for academy means: U17 ~1.03s, U19 ~1.03s, U21 ~1.02s.

Thresholds (lower is better)

Age Group	Poor	Average	Good	Elite
U-17	>1.15 s	1.08-1.15 s	1.02-1.07 s	<1.02 s
U-19	>1.12 s	1.05-1.12 s	0.99-1.04 s	<0.99 s
U-21	>1.10 s	1.03-1.10 s	0.97-1.02 s	<0.97 s

Confidence: MEDIUM — The Austrian NLZ (Gonaus) and Hoffenheim (Ruf) datasets are strong, but 5m is not universally tested across European academies. The narrow time window (fractions of a second) makes threshold

placement more sensitive to measurement error.

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4. 10m Sprint

Published Data

Study	N	Age Group	Level	Mean +/- SD (s)
Gonaus et al. (2019)	261-998 per age	U15-U19	Austrian Bundesliga NLZ	Age-specific norms
Bujnovsky et al. (2019)	—	U17	Czech First League	~1.85-1.90
Mendez-Villanueva & Buchheit (2011)	—	U13-U18	Aspire Academy (Qatar)	Age-specific data

Threshold Derivation

The Gonaus et al. (2019) Austrian NLZ dataset is the most powerful source for 10m sprint, with hundreds of athletes per age group across multiple seasons. This was the primary source for threshold derivation. Czech First League (Bujnovsky) and Aspire Academy (Mendez-Villanueva/Buchheit) data served as cross-references.

Best estimates for academy means: U17 ~1.77s, U19 ~1.75s, U21 ~1.74s.

Thresholds (lower is better)

Age Group	Poor	Average	Good	Elite
U-17	>1.88 s	1.80-1.88 s	1.74-1.79 s	<1.74 s
U-19	>1.84 s	1.76-1.84 s	1.70-1.75 s	<1.70 s
U-21	>1.82 s	1.74-1.82 s	1.68-1.73 s	<1.68 s

Confidence: HIGH — The Austrian NLZ dataset (Gonaus et al.) provides robust, large-sample norms. 10m sprint is one of the most commonly tested distances in European football academies, and multiple studies converge on similar values.

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5. 30m Sprint

Published Data

Study	N	Age Group	Level	Mean +/- SD (s)
Nowak et al. (2025)	—	Age 15 (P50)	Polish academy	4.268
Nowak et al. (2025)	—	Age 16 (P50)	Polish academy	4.220
Mikulic et al. (2025)	—	U17-U19	Slovak elite academy	Age-specific data
Ruf et al. (2024)	—	U11-U19	TSG Hoffenheim (Bundesliga NLZ)	30m centile data

Threshold Derivation

The Ruf et al. (2024) Hoffenheim dataset is particularly valuable as a direct Bundesliga NLZ source, providing centile-based norms for 30m sprint from U11 through U19. The Polish (Nowak) and Slovak (Mikulic) data provide additional European reference points.

Best estimates for academy means: U17 ~4.22s, U19 ~4.18s, U21 ~4.15s.

Thresholds (lower is better)

Age Group	Poor	Average	Good	Elite
U-17	>4.50 s	4.30-4.50 s	4.15-4.29 s	<4.15 s
U-19	>4.40 s	4.20-4.40 s	4.05-4.19 s	<4.05 s
U-21	>4.35 s	4.15-4.35 s	4.00-4.14 s	<4.00 s

Confidence: MEDIUM-HIGH — The Hoffenheim Bundesliga NLZ data (Ruf et al.) is directly relevant. Polish and Slovak supporting data are consistent. However, the total number of large-sample studies for 30m is fewer than for 10m or CMJ.

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6. 30-15 Intermittent Fitness Test (IFT) — VIFT

Published Data

Study	N	Age Group	Level	Mean +/- SD (km/h)
Buchheit et al. (2013)	—	U16	Aspire Academy	19.2 +/- 0.9
Buchheit et al. (2013)	—	U18	Aspire Academy	19.7 +/- 1.0
Fernandez Javega & Pena-Gonzalez (2024)	—	U15	Spanish youth	18.99 +/- 1.31
Fernandez Javega & Pena-Gonzalez (2024)	—	U16	Spanish youth	18.81 +/- 1.55
Silva et al. (2022)	—	U14-U19	Turkish youth	17.44-19.54 (progressive)
Buchheit (2021)	—	Elite male	General benchmark	~19.5

Threshold Derivation

The 30-15 IFT is widely used in European football but has less published normative data specific to German NLZ than physical performance tests. No German Bundesliga NLZ-specific 30-15 IFT data exists in the literature, as the DFB uses a different testing battery for its academies.

The Buchheit et al. (2013) Aspire Academy data is the most cited reference. Spanish (Fernandez Javega & Pena-Gonzalez) and Turkish (Silva) studies provide supporting age-group data. Buchheit's (2021) general benchmark of ~19.5 km/h for elite males provides an anchor point.

Thresholds reflect the progressive improvement from U-17 to U-21, with the elite threshold set at levels consistent with professional or near-professional aerobic fitness.

Thresholds (higher is better)

Age Group	Poor	Average	Good	Elite
U-17	<17.5 km/h	17.5-19.0 km/h	19.1-20.5 km/h	>20.5 km/h
U-19	<18.5 km/h	18.5-20.0 km/h	20.1-21.5 km/h	>21.5 km/h
U-21	<19.0 km/h	19.0-20.5 km/h	20.6-22.0 km/h	>22.0 km/h

Confidence: MEDIUM — Good reference data from Aspire Academy and multiple European studies, but no German NLZ-specific norms. The 30-15 IFT protocol is well-standardized, which improves cross-study comparability, but club-specific training loads and testing conditions can influence results.

Confidence Summary

Test	Confidence	Rationale
CMJ	**HIGH**	Hoppe et al. (2020) German NLZ n=1,134; corroborated by 4+ European studies
Broad Jump	**MEDIUM / LOW**	Limited football-specific data; U-19/U-21 partially estimated
5m Sprint	**MEDIUM**	Austrian NLZ + Hoffenheim data, but 5m not universally tested
10m Sprint	**HIGH**	Gonaus et al. Austrian NLZ (hundreds per age); widely tested distance
30m Sprint	**MEDIUM-HIGH**	Hoffenheim Bundesliga NLZ centiles + Polish/Slovak supporting data
30-15 IFT	**MEDIUM**	Aspire Academy + Spanish/Turkish references; no German NLZ data

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Full Citations

Primary Sources

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11. **Deprez, D., Fransen, J., Boone, J., Lenoir, M., Philippaerts, R., & Vaeyens, R. (2015).** Characteristics of high-level youth soccer players: variation by playing position. *Journal of Sports Sciences*, 33(3), 243-254. https://doi.org/10.1080/02640414.2014.934707

12. **Bujnovsky, D., Maly, T., Ford, K. R., Sugimoto, D., Kunzmann, E., Hank, M., & Zahalka, F. (2019).** Physical fitness characteristics of high-level youth football players: influence of playing position. *Sports*, 7(2), 46. https://doi.org/10.3390/sports7020046

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Notes for Future Updates

- **Priority improvement:** Obtain Broad Jump data from German Bundesliga NLZ to strengthen the currently low-confidence U-19/U-21 thresholds.
- **30-15 IFT:** If 1. FC Koln ITP begins collecting 30-15 IFT data systematically, internal norms should eventually supplement published benchmarks.
- **Custom thresholds:** The ITP Player Database supports coach-customized thresholds that override these defaults. Coaches should adjust based on their own squad data once sufficient testing sessions have been collected.
- **Measurement standardization:** Thresholds assume standardized testing conditions (indoor timing gates for sprints, force platform or contact mat for CMJ). Non-standard setups may introduce systematic bias.